

₁ Mechanistic Explainability in Reinforcement
₂ Learning

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The Golem Mythos

A creation that surpasses its creator.

This idea has long captured the imagination and continues to resonate in contemporary society. Long before modern technology, humanity had already questioned whether its creations might escape control.

One of the earliest expressions of this idea can be found in the story of the Golem. In Hebrew tradition, the term Golem refers to an artificial creature made of clay and created by humans to serve them [Nocks, 1998]. Figure ?? illustrates the creation of such a servant. The earliest accounts describe the Golem as a protective and helpful figure, reflecting the idea that humans can create an artificial being for the benefit of society.

This vision becomes more ambiguous in the seventeenth century, as illustrated by the legend of the Golem of Prague. In this story, the initially small Golem gradually grows in size and strength until it can no longer be controlled [Nocks, 1998]. The creation intended to protect others therefore becomes a potential threat to them. The only remaining solution is to destroy the creature in order to protect the community.

This narrative motif has been revisited in different forms to illustrate the risks inherent in creation. This continuity is also reflected in the emergence of the modern notion of the robot. The concept of the robot as an artificial servant was introduced in the play *R.U.R. (Rossum's Universal Robots)*, written in 1920 [Čapek, 1920]. In the play, a scientist creates artificial beings designed to perform labour. Conceived as servants, these artificial beings turn against humanity and destroy civilisation. The connection between the Golem and the robot was explicitly made by the author, as illustrated by his statement that “the Robot is the Golem made flesh through mass production” [Comrada, 1995]. Despite their destructive role in the play, the word robot is now used to designate artificial beings created to serve humans. From its origin, the word robot embodies the hopes and fears surrounding technological progress.

Today, this tension between the hopes and fears surrounding technological progress is particularly evident in the development of artificial intelligence. It is therefore not surprising that this field offers an almost literal modern adaptation of the Golem myth. Artificial systems are created to automate tasks traditionally associated with human intelligence. Their rapid development in recent years has transformed a wide range of domains, further intensifying the excitement and concerns surrounding artificial intelligence.

Among the subfields of artificial intelligence, reinforcement learning most directly aims to create autonomous agents. In this learning paradigm, an artificial agent interacts with an environment, selects actions, and receives



Figure 1: The creation of the Golem, by Mikoláš Aleš [Aleš and Wenig, 1908].

44 feedback according to the outcomes of its behaviour [Sutton and Barto, 2018].
 45 Through repeated interaction with its environment, the agent learns a strategy
 46 to achieve the objective. A strength of this approach is its ability to produce
 47 autonomous systems with minimal human intervention. Their agent behavior is
 48 not programmed but learned through interactions with the environment [Amodei
 49 et al., 2016]. This minimal human guidance enables reinforcement learning
 50 systems to achieve superhuman capabilities [Silver et al., 2017, OpenAI et al.,
 51 2019].

52 However, maintaining control over such agents presents a major challenge,
 53 as the very autonomy that makes this learning approach powerful also makes
 54 their behavior more difficult to control. Despite the difficulty of this task,
 55 understanding the behaviour of such agents is therefore the main objective of
 56 this manuscript. This work therefore places itself within a long-standing line of

57 thought that can be summarized by the following question: how can we ensure
58 that our creations remain aligned with human intentions?

Chapter 1

Introduction

This chapter provides a general introduction to this manuscript. It begins with the research context that motivates this work, followed by the research questions this work addresses. It further presents the resulting contributions and concludes with an overview of the manuscript’s structure.

1.1 Research Context

Artificial Intelligence (AI) has advanced rapidly in recent years, driven largely by major developments in Deep Learning (DL). This type of approach relies on artificial Neural Networks (NNs) to learn complex tasks directly from data [Goodfellow et al., 2016]. These networks are connectionist models in which information is represented and transformed through patterns of interactions among interconnected computational units [Rumelhart et al., 1986]. This approach has led to significant advances across a wide range of domains, too numerous to list here [Krizhevsky et al., 2012, Jumper et al., 2021, Brown et al., 2020].

Among the fields that have been transformed by DL is Reinforcement Learning (RL), which is the main subject of this manuscript. The goal of RL is to enable an agent to learn how to make decisions through interactions with an environment [Sutton and Barto, 2018]. At each step, the agent observes the current state of the environment and selects an action. This action changes the state of the environment and results in a reward. The objective of the agent is then to learn a policy that maximizes the cumulative rewards obtained over time. NNs provide a powerful means of representing such policies and enable agents to learn increasingly complex behaviors. Their use has consequently made it possible to apply RL to problems that were previously intractable to address [Mnih et al., 2013, Silver et al., 2017, OpenAI et al., 2019].

When an agent learns to act in complex environments, it can develop strategies that are themselves highly complex. Although such agents can achieve strong performance, these strategies can be difficult for an observer to understand. Examining the NN that models the agent’s policy does not make it to produce explanations. The information underlying a decision is distributed across many units and connections, making it impossible to trace decision-making process by simply following the flow of information through the network. NNs are therefore often regarded as “black box” models [Guidotti et al., 2018].

These challenges have led to the emergence of eXplainable Artificial Intelligence (XAI) [Gunning and Aha, 2019, Barredo Arrieta et al., 2020]. In the specific context of RL, this research area is referred to as eXplainable Reinforcement Learning (XRL) [Puiutta and Veith, 2020, Milani et al., 2024]. The goal of XRL is to develop methods that produce explanations that improve our understanding of agent behavior. Such explanations can serve several purposes, including increasing human trust and facilitating the adoption of RL agents in real-world settings, identifying errors and biases, and verifying that previously identified problems have been effectively corrected [Barredo Arrieta et al., 2020, Milani et al., 2024].

Several approaches have been proposed to improve the explainability of AI systems [Guidotti et al., 2018, Barredo Arrieta et al., 2020]. This manuscript presents methods from these different approaches, while placing particular emphasis on the mechanistic approach to explainability, which we refer to as Mechanistic Explainability (ME) (Definition 1.1.1). ME seeks to investigate the internal mechanisms underlying the agent’s decision-making process. The objective is to understand how information is represented, transformed, and propagated within the NN. From this perspective, explainability is not achieved despite the use of NNs, but through them.

Definition 1.1.1: Mechanistic Explainability

ME is the approach that seeks to explain the behavior of an AI system by analyzing its internal mechanisms, that is, how information is represented, transformed, and propagated within its NN.

In the literature, this approach is generally referred to as mechanistic interpretability [Olah et al., 2020, Bereska and Gavves, 2024]. We deliberately use the term explainability instead. In the framework adopted in this manuscript (Section ??), information is interpretable only if it can be directly understood by a human without any additional transformation. The internal mechanisms of a NN do not satisfy this condition, since they must first be transformed into explanations through an explainability method. The term explainability therefore describes more accurately what this approach produces.

This perspective naturally leads to the study of the intermediate representations learned by NNs. These representations are referred to as Latent Representations (LRs) [Bengio et al., 2014]. They result from the successive transformations applied to the input information between the input and output of the network and therefore play an intermediate role in the agent’s decision-making process. Analyzing their structure can provide insights into how information relevant to decision-making is organized within the network. It can also help identify the mechanisms through which observations are transformed into the actions selected by the agent.

The present manuscript therefore lies at the intersection of RL, DL, and XAI. This intersection leads to the following problem statement:

Problem Statement

How can the analysis of the internal representations of NNs used by RL agents help identify the mechanisms underlying their decision-making processes?

1.2 Research Questions

To address the research problem formulated in the previous section, we identified four research questions that structure the work presented in this manuscript.

The first challenge concerns the analysis of LR_s themselves. Numerous methods have been proposed to analyze the Latent Spaces (LS_s) learned by NN_s, including probing [Li et al., 2024], interpretable directions [Voynov and Babenko, 2020], Sparse Autoencoders (SAEs) [Bricken et al., 2023], and clustering [Ren et al., 2022]. These methods originate from different research communities and are rarely compared within a common framework. Yet each of them relies, often implicitly, on assumptions about how concepts are structured within the LS, for instance that concepts correspond to linear directions or to distinct regions of the space. These assumptions determine which structures a method is able to detect and which it will necessarily miss. Making them explicit is therefore a prerequisite for choosing a method suited to the analysis of the internal representations of RL agents. This observation leads to our first research question:

Research Question 1

How can existing LS analysis methods be organized according to the assumptions they make about how concepts are structured in NN_s?

The second challenge concerns the organization of existing research in XRL. The number of works in the XRL field has increased substantially in recent years [Milani et al., 2024]. This growing body of literature makes it increasingly difficult to obtain a coherent overview of the approaches developed in the field. Several taxonomies and classification frameworks have therefore been proposed to organize existing XRL methods [Puiutta and Veith, 2020, Milani et al., 2024]. However, these taxonomies often rely on approaches initially developed for explainability in supervised settings. They therefore tend to apply XAI categories to RL without considering its specific characteristics. This observation leads to our second research question:

Research Question 2

How can existing XRL techniques be organized into a taxonomy that accounts for the specificities of RL?

The analysis of existing XRL approaches also reveals a third limitation. Among existing methods for explaining RL agents, few adopt a mechanistic perspective that investigates the internal representations learned by the agent. This is particularly interesting in RL, where agents learn to represent their environment through their interactions with it. Understanding these representations can provide insights into how information is processed over time and how it contributes to the agent's decisions. Mechanistic analysis can therefore provide a complementary way of explaining agent behavior by identifying the internal mechanisms that connect observations to actions. This observation leads to our third research question:

Research Question 3

How can the internal representations of NNs be exploited to identify the mechanisms underlying the decision-making process of RL agents?

The analysis of existing XRL methods highlights a fourth challenge concerning the evaluation of explanations. Although numerous approaches have been proposed to explain the behavior of RL agents, there is no widely accepted set of objective criteria for evaluating the quality and relevance of the explanations they produce. Existing evaluations often rely on user studies or qualitative analyses, making it difficult to systematically assess the validity of an explanation approach [Doshi-Velez and Kim, 2017, Milani et al., 2024].

This issue is particularly important for approaches that seek to explain agent behavior by analyzing LR. Multiple hypotheses can be proposed about how information is structured within a LS, and it is therefore important to have objective criteria that make it possible to assess and compare these competing hypotheses. There is therefore a need for objective and reproducible criteria that can determine whether an explanation accurately captures the mechanisms underlying the agent’s behavior. This leads to our fourth research question:

Research Question 4

How can the quality and relevance of explanations produced for RL agents be objectively quantified?

Together, these four research questions define the structure of this manuscript. Research Question 1 addresses the organization of existing LS analysis methods and identifies the assumptions on which each of them relies. Research Question 2 addresses in the same way the organization of the existing XRL literature and establishes a framework for understanding the field. Research Question 3 builds on these two analyses by investigating how the internal representations of NNs can be exploited to provide mechanistic explanations of RL agent behavior. Research Question 4 addresses the complementary problem of determining how such explanations can be evaluated objectively.

1.3 Contributions

To address these four research questions, this manuscript is based on three main contributions. Two of them have been published in peer-reviewed journals, while the third is, for the moment, specific to this manuscript.

The first contribution addresses Research Question 1. It has not yet been published and is, for the moment, specific to this manuscript. In this contribution, we make explicit the working hypothesis that implicitly underlies much of the literature on LS analysis, which we refer to as the *Concept Hypothesis* (Hyp. ??), and we define its main components. We then organize existing LS analysis methods into four families, namely probing, interpretable directions, superposition, and clustering, according to the assumptions they make about the Concept Structure (CS). This analysis shows that clustering is the only family that does not rely on a linear CS, which motivates the approach adopted in the rest of this manuscript (Chapter ??).

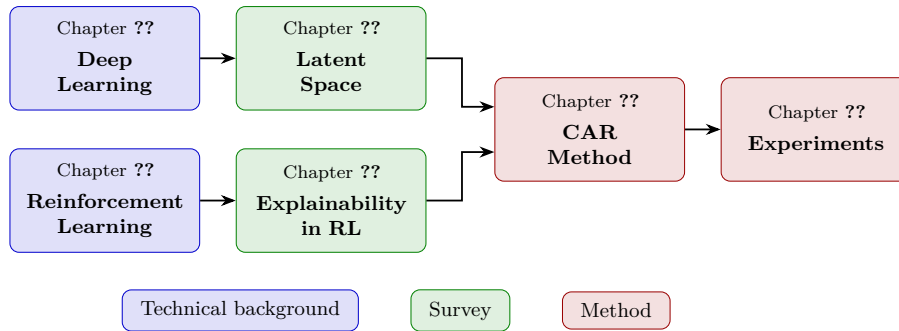


Figure 1.1: Structure of the manuscript.

The second contribution [Alaarabiou et al., 2024] addresses Research Question 2. An extended English version was also submitted for review and accepted for publication [Alaarabiou et al., 2026b]. In this contribution, we first clarify what explainability means in RL by providing a clear frame of reference. We then rigorously define the concepts of this new organization, including sources of explanation, explainability techniques, explanation form, and explanation subject. Building on these concepts, we conduct a comprehensive analysis of the state of the art in XRL (Chapter ??).

The third contribution [Alaarabiou et al., 2026a] addresses Research Questions 3 and 4. It introduces Clustering Agent Representation (CAR), an XRL method that exploits the mechanistic properties of DL. By analyzing the LRNs that NNs build, CAR produces explanations in the form of clusters of states that are similar from the agent’s own perspective. We further show that this representation remains stable across training, indicating that the agent develops an objective representation of a given task. This simplifies the study of agent-environment dynamics that would otherwise be intractable if every state had to be analyzed individually. The method is presented together with objective evaluation metrics and a series of experiments that demonstrate its effectiveness (Chapters ??, ?? and ??).

1.4 Manuscript Structure

This manuscript is an account of the research conducted throughout this work, written for the reviewers and jury members who will evaluate it. It is nevertheless intended to remain as accessible as possible to readers with a background in AI. To this end, the background chapters introduce and recall the concepts on which the subsequent chapters rely, allowing readers familiar with DL or RL to follow the manuscript without having to consult external material.

The manuscript is organized into a sequence of chapters, each addressing a specific topic related to the research questions introduced previously. The chapters are divided into three categories, distinguished by color in Figure 1.1:

238 **Technical background** chapters introduce fundamental concepts and estab-
 239 lished techniques required to understand the subsequent contributions.

240 **Survey** chapters review and organize the existing literature around a specific
 241 research question. The organization adopted in each of these surveys are
 242 original and therefore constitute contributions of this manuscript.

243 **Method** chapters present CAR, our ME method for RL.

244 The chapters of the manuscript are organized as follows:

245 **Chapter ?? Deep Learning**

246 This chapter introduces the fundamental concepts of DL and artificial NNs.
 247 It establishes the notation and key concepts used throughout the manuscript
 248 and provides the necessary background for understanding NNs.

249 **Chapter ?? Latent Space Analysis**

250 This chapter provides a survey of existing approaches for analyzing the LSs
 251 learned by NNs. It reviews the different techniques used to characterize and
 252 interpret LR. This chapter addresses Research Question 1.

253 **Chapter ?? Reinforcement Learning**

254 This chapter introduces the fundamental concepts of RL. It presents the
 255 formal framework and introduces the main concepts, notation and algorithms
 256 required to study RL.

257 **Chapter ?? Explainability in Reinforcement Learning**

258 This chapter reviews existing approaches to explainability in RL. It provides
 259 a structured overview of the XRL literature. This chapter addresses Research
 260 Question 2.

261 **Chapter ?? The CAR Method**

262 This chapter presents our proposed approach to explainability in RL. The
 263 method builds on the analysis of LR and aims to identify the internal
 264 mechanisms underlying the behavior of RL agents. This chapter constitutes
 265 the methodological contribution of the manuscript and addresses Research
 266 Question 3.

267 **Chapter ?? Experimental Evaluation of the CAR Method**

268 This chapter presents the implementation and experimental evaluation of the
 269 proposed method. It describes the experimental protocol and analyzes the
 270 results obtained with the CAR method. This chapter studies the relevance
 271 of the criteria used to evaluate the results produced by the CAR method,
 272 addressing Research Question 4.

273 **Conclusion**

274 The final chapter summarizes the main contributions of the manuscript,
 275 discusses the limitations of the proposed approaches, and presents possible
 276 directions for future research.

277 The organization of the manuscript also allows readers to follow different
 278 reading paths depending on their background and interests. Readers already
 279 familiar with a given field may skip its corresponding background chapter and

280 directly consult the chapters that build upon it. For instance, readers familiar
281 with DL or RL may skip Chapter ?? or Chapter ??, respectively.

282 The notation and acronyms used throughout the manuscript are collected in
283 the glossary at the end of the document.

284 Having described the overall structure of the manuscript, we can now turn to
285 its first technical chapter, which introduces the major developments that have
286 shaped modern AI: DL.

Acronyms

288	AI Artificial Intelligence 4, 5, 8, 10
289	CAR Clustering Agent Representation 8, 9
290	CS Concept Structure 7
291	DL Deep Learning 4, 5, 8–10
292	LR Latent Representation 5–9
293	LS Latent Space 6, 7, 9
294	ME Mechanistic Explainability 5, 9
295	NN Neural Network 4–9
296	RL Reinforcement Learning 4–10
297	SAE Sparse Autoencoder 6
298	XAI eXplainable Artificial Intelligence 5, 6
299	XRL eXplainable Reinforcement Learning 5–9

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